

Swinburne University of Technology

COS30045 - Data Visualisation

Lab Exercise Demonstration 3 (Exercises 5.1 - 6.2)

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Lab 6.1

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8" />
  <meta name="description" content="Data Visualisation Exercise 1" />
  <meta name="keywords" content="HTML, CSS" />
  <meta name="author" content="Your name here" />

  <title>Task 6.1 D3 Interactivity - Mouse Over Effects</title>
  <script src="https://d3js.org/d3.v7.min.js"></script>

  <style>
    button {
      border: none;
      color: white;
      padding: 10px 20px;
      text-align: center;
      text-decoration: none;
      display: inline-block;
      font-size: 16px;
      margin: 10px 5px;
      cursor: pointer;
      border-radius: 5px;
      transition-duration: 0.4s;
    }

    #add {
      background-color: #4CAF50;
    }

    #add:hover {
      background-color: #45a049;
    }

    #remove {
      background-color: #2196F3;
    }

    #remove:hover {
      background-color: #0b7dda;
    }
  </style>
</html>
```

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    svg {
      margin-top: 20px;
    }
  </style>

</head>

<body>
  <h1>Bar Chart with Mouse Over</h1>
  <button id="add">Add</button>
  <button id="remove">Remove</button>
  <br>

  <script>
    // example from Murray
    var dataset = [20, 5, 26, 23, 9, 23, 16, 4, 29];
    var w = 500;
    var h = 150;
    var padding = 1;
    var maxValue = 25;

    var xScale = d3.scaleBand()
      .domain(d3.range(dataset.length))
      .rangeRound([0, w])
      .paddingInner(0.05);

    var yScale = d3.scaleLinear()
      .domain([0, d3.max(dataset, function (d) {
        return d;
      })]) // The data input range
      .range([0, 150]); // The range available for visualization on
screen

    var svg1 = d3.select("body")
      .append("svg")
      .attr("width", w)
      .attr("height", h);

    //initial bars
    svg1.selectAll("rect")
      .data(dataset)
      .enter()
      .append("rect")
      .attr("x", function (d, i) {

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        return xScale(i);
    })
    .attr("y", function (d) {
        return h - yScale(d);
    })
    .attr("width", xScale.bandwidth())
    .attr("height", function (d) {
        return yScale(d);
    })
    .attr("fill", "teal")
    .on("mouseover", function (event, d) {

        // Get the position of the bar
        var xPosition = parseFloat(d3.select(this).attr("x")) +
xScale.bandwidth() / 2;
        var yPosition = parseFloat(d3.select(this).attr("y")) + 15; //
Move tooltip above the bar

        // Add new tooltip to the position
        svg1.append("text")
            .attr("id", "tooltip")
            .attr("x", xPosition)
            .attr("y", yPosition)
            .attr("text-anchor", "middle")
            .attr("font-family", "sans-serif")
            .attr("font-size", "13px")
            .attr("fill", "black")
            .text(d);

        //change bar to orange
        d3.select(this)
            .transition().duration(200)
            .attr("fill", "orange");

    })
    .on("mouseout", function () {
        // Remove the tooltip when the mouse moves out
        d3.select("#tooltip").remove();

        //back to teal
        d3.select(this)
            .transition().duration(200)
            .attr("fill", "teal");
    });

```

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//update chart after button clicked
function updateChart() {
    xScale.domain(d3.range(dataset.length))

    // Update the Y scale based on new data
    yScale.domain([0, d3.max(dataset)]);

    //update bars
    var bars = svg1.selectAll("rect")
        .data(dataset);

    //add bars
    bars.enter()
        .append("rect")
        .attr("x", w)
        .attr("y", function (d) {
            return h - yScale(d);
        })
        .attr("fill", "teal")
        .merge(bars)
        .transition()
        .duration(200)
        .attr("x", function (d, i) {
            return xScale(i);
        })
        .attr("y", function (d) {
            return h - yScale(d);
        })
        .attr("width", xScale.bandwidth())
        .attr("height", function (d) {
            return yScale(d);
        });

    // Add mouseover and mouseout effect to both old and new bars
    svg1.selectAll("rect")
        .on("mouseover", function (event, d) {
            // Get the position of the bar
            var xPos = parseFloat(d3.select(this).attr("x")) +
xScale.bandwidth() / 2;
            var yPos = parseFloat(d3.select(this).attr("y")) + 15;
            // Move tooltip above the bar

            // Add new tooltip

```

```

        svg1.append("text")
            .attr("id", "tooltip")
            .attr("x", xPos)
            .attr("y", yPos)
            .attr("text-anchor", "middle")
            .attr("font-family", "sans-serif")
            .attr("font-size", "13px")
            .attr("fill", "black")
            .text(d);

        d3.select(this)
            .transition().duration(200)
            .attr("fill", "orange");
    })
    .on("mouseout", function () {
        // Remove the tooltip when the mouse moves out
        d3.select("#tooltip").remove();

        //back to teal
        d3.select(this)
            .transition().duration(200)
            .attr("fill", "teal");
    });

    // Exit bars
    bars.exit()
        .transition()
        .duration(200)
        .attr("x", w)
        .remove();
}

// Add new data button
d3.select("#add").on("click", function () {

    var newNumber = Math.floor(Math.random() * maxValue);
    dataset.push(newNumber);
    updateChart();
});

// Remove data button
d3.select("#remove").on("click", function () {
    dataset.shift();
    updateChart();
});

```

```

});

</script>

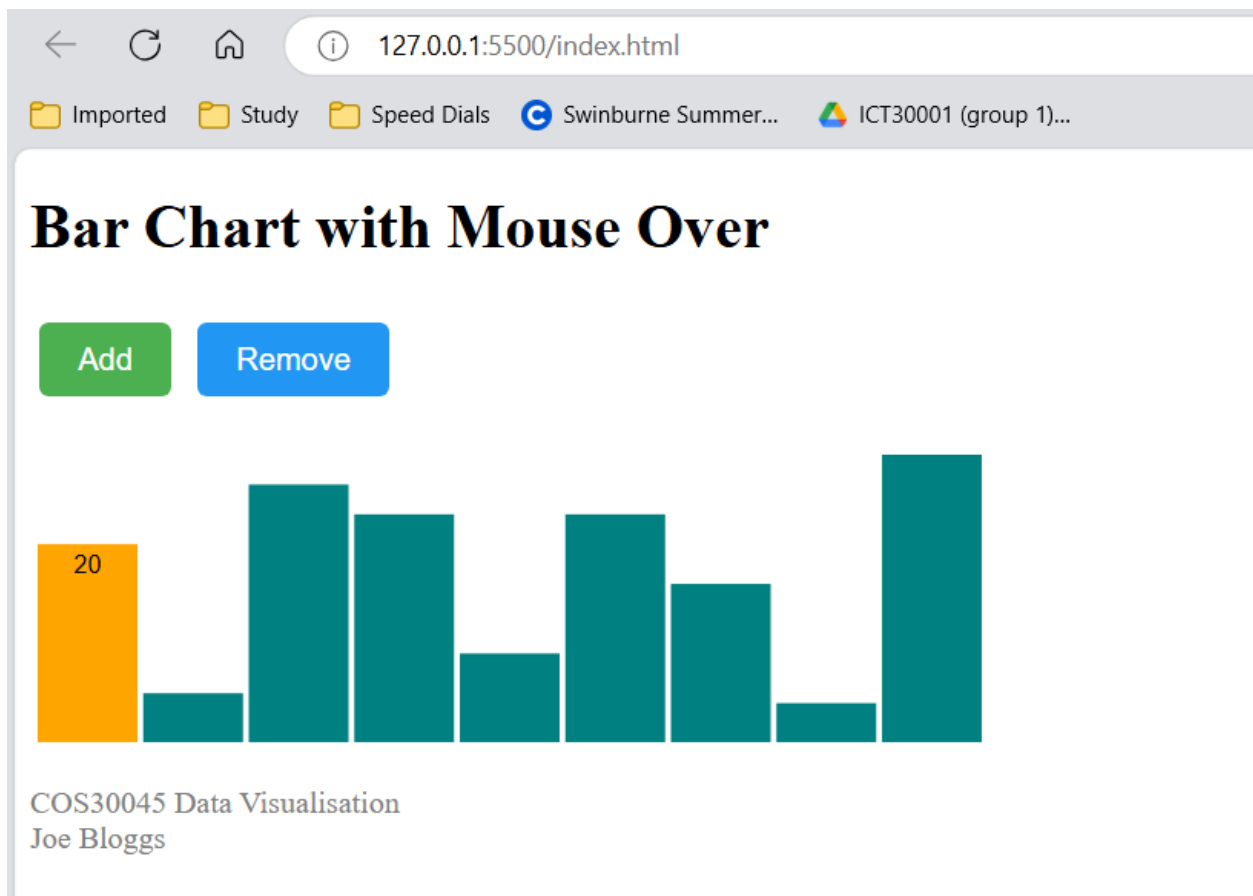
<br><br>
<footer style="color:grey">COS30045 Data Visualisation<br>Joe Bloggs</footer>

</body>

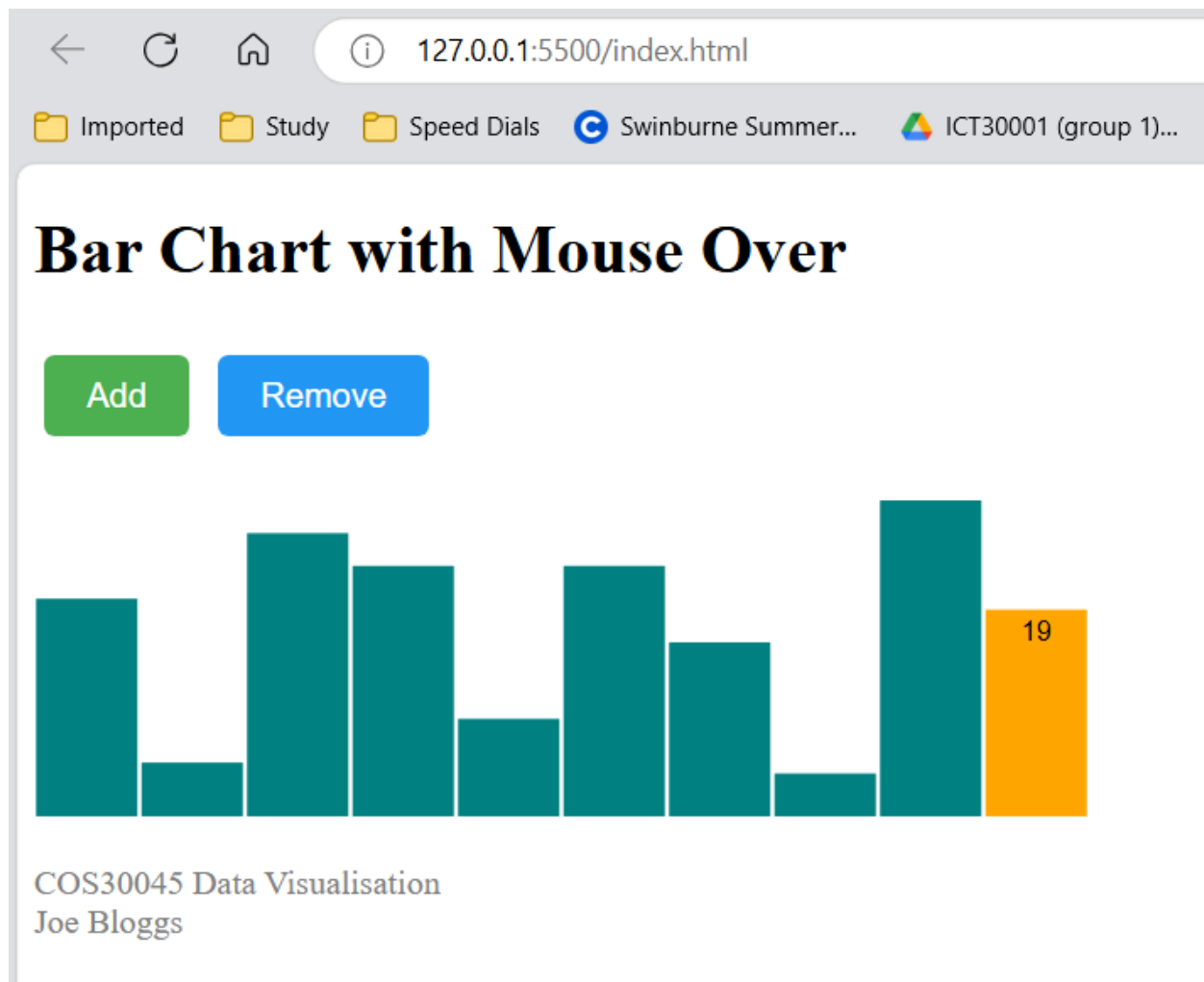
</html>

```

Initial bar chart with the first bar being hovered:



This is when a new bar is added and hovered:



This is when the first bar (20) is removed:

Bar Chart with Mouse Over

Add

Remove



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